CASE STUDY

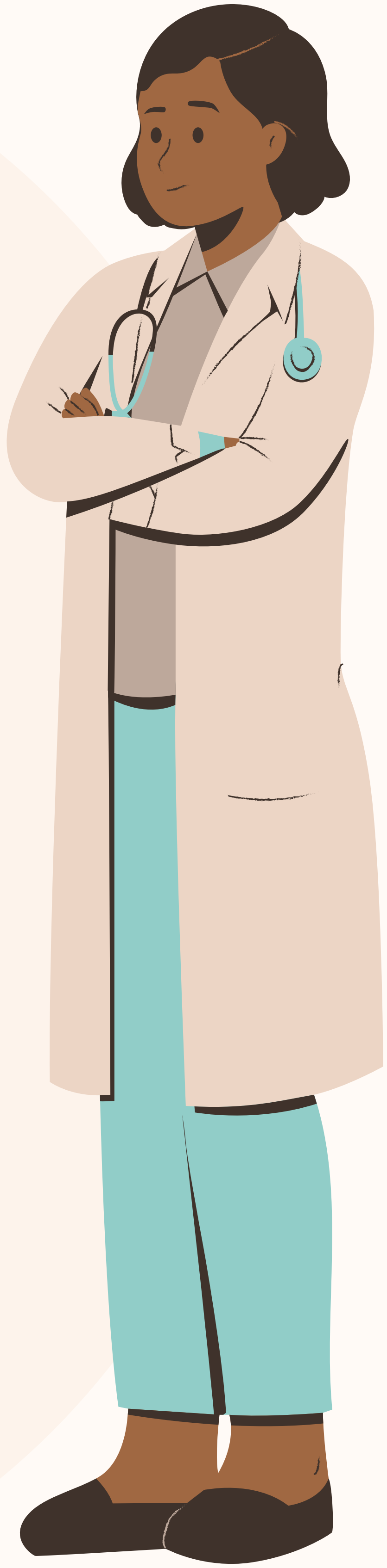
HOSPITAL MANAGEMENT SYSTEM

INTRODUCTION

A Hospital Management System (HMS) is a software application designed to manage and organize hospital operations digitally. It helps hospitals and clinics maintain patient records, doctor details, treatments, appointments, and reports in an efficient and secure manner.

Traditionally, hospitals used paper-based record systems, which were time-consuming, difficult to maintain, and prone to errors or data loss. The Hospital Management System solves these problems by storing information digitally and providing quick access to hospital records whenever required
.

This project is developed using Python and focuses on simplifying hospital management tasks such as adding patient details, searching records, viewing patient information, and generating reports. The system improves accuracy, reduces paperwork, saves time, and enhances the overall efficiency of hospital operations. The Hospital Management System is useful for hospitals, clinics, and healthcare centers to ensure better patient care and organized data management.



🎯 Objectives of Hospital Management System

1)To manage patient records digitally
Store and maintain patient information in an organized digital format.

2)To reduce manual paperwork
Minimize the use of paper records and manual data handling in hospitals.

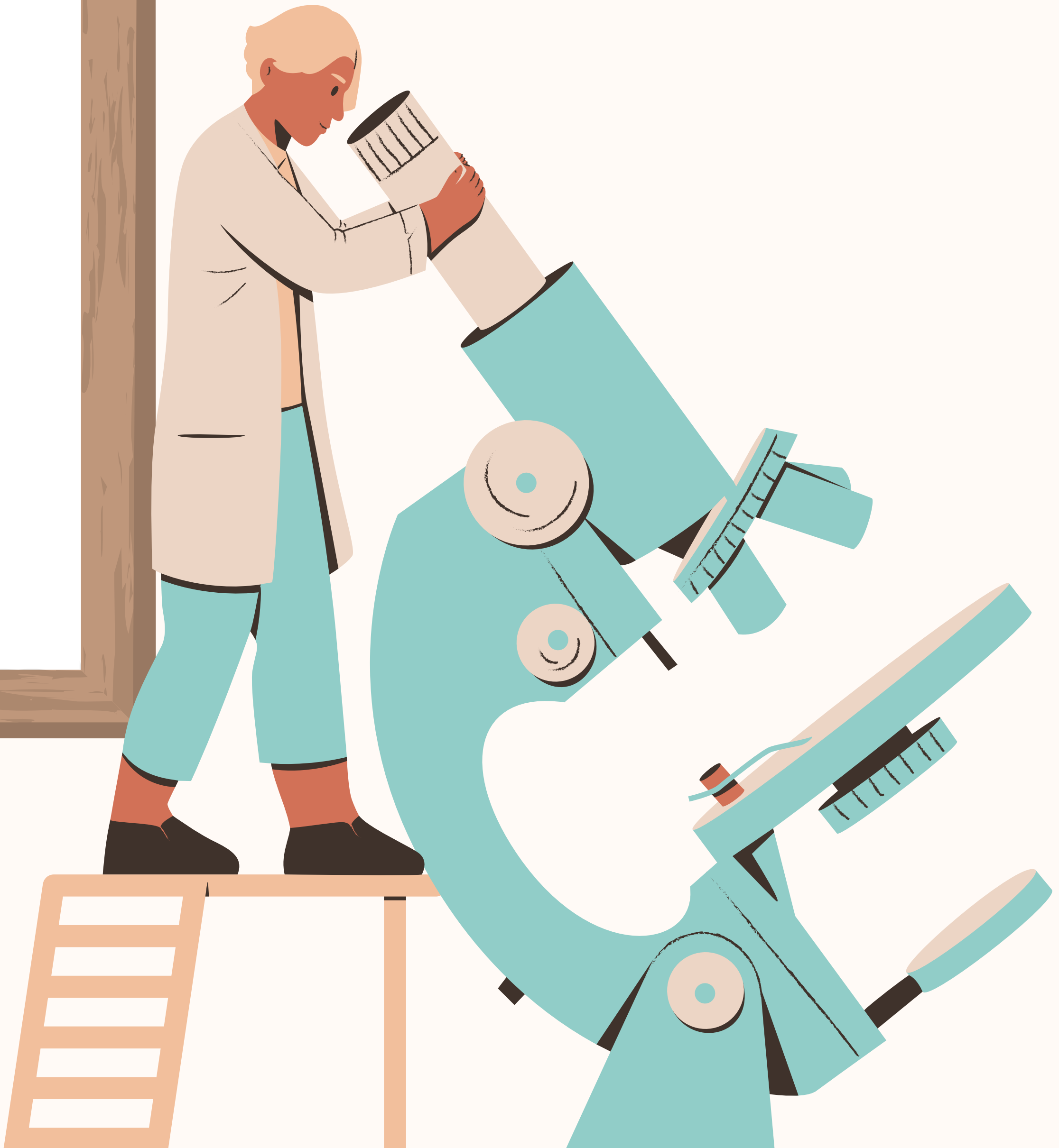
3)To improve data organization
Keep doctor, patient, treatment, and appointment details properly structured.

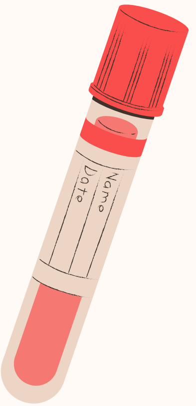
4)To provide quick access to records
Enable fast searching, viewing, and updating of hospital information.

5)To increase efficiency in hospital operations
Save time for hospital staff and improve workflow management.

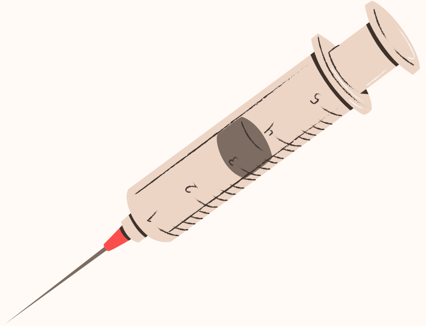
OBJECTIVES

For Better Life





Importance Of Hms



Overview 01

Reduces paperwork by storing records digitally.

Saves time in managing patient and doctor information.

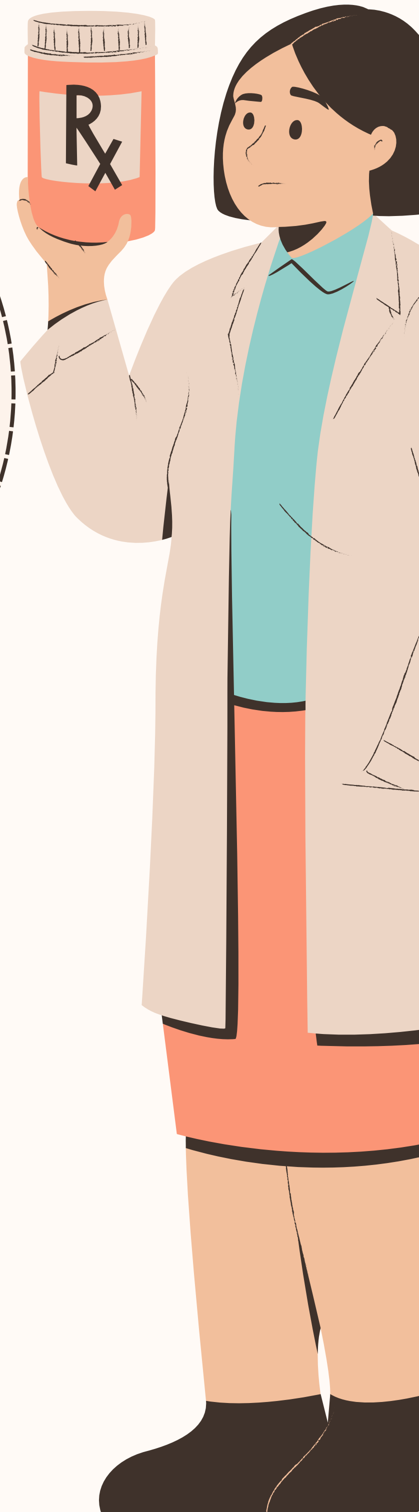
Provides quick access to patient records and reports.

Overview 02

Improves accuracy and reduces human errors.

Prevents data loss caused by damaged or misplaced paper files.

Helps manage appointments and treatments efficiently.



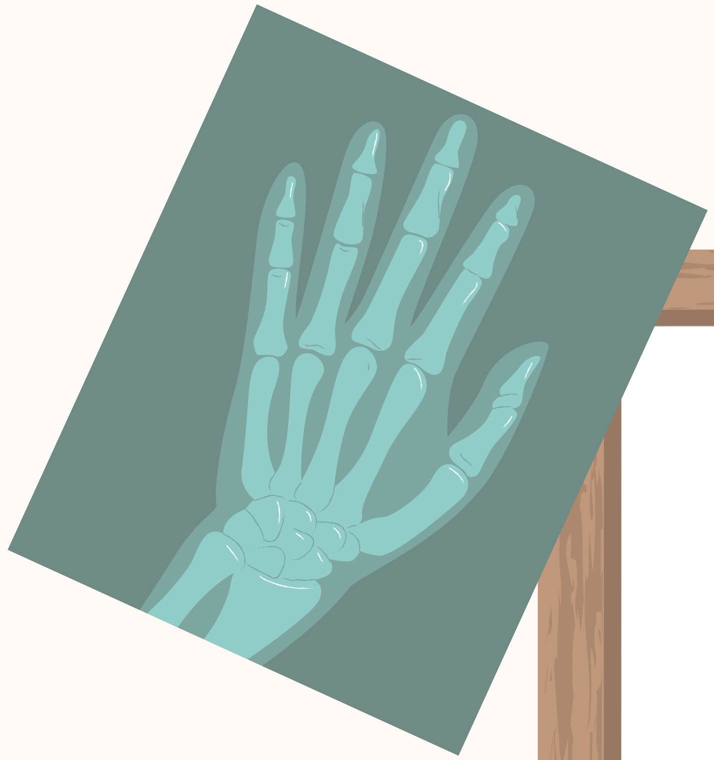
Overview 03

Laboratory Management

Data Security and Confidentiality

Reporting and Decision Making

Working Of Hms



Patient Registration

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Appointment Booking

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Doctor Consultation

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Lab Tests / Pharmacy

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Billing & Payment

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Admission/Discharge (if needed)

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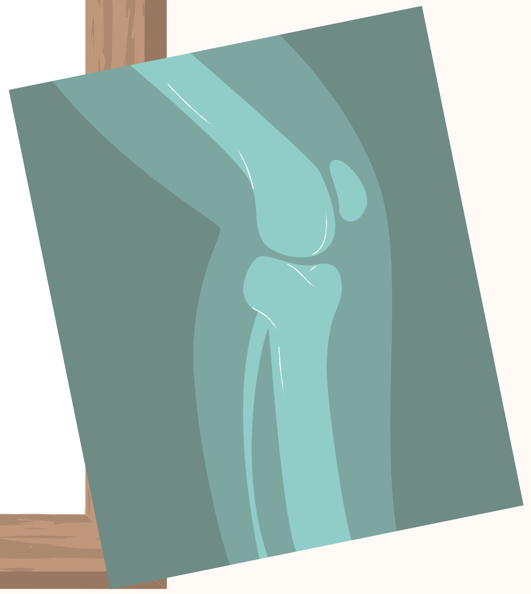
Report Generation & Record Storage



CODE

```python
# Hospital Management System in Python
patients = []

def menu():
    print("\n===== HOSPITAL MANAGEMENT SYSTEM =====")
    print("1. Add Patient")
    print("2. View Patients")
    print("3. Search Patient")
    print("4. Remove Patient")
    print("5. Exit")

while True:
    menu()

    choice = input("Enter your choice: ")

    # Add Patient
    if choice == "1":
        name = input("Enter patient name: ")
        age = input("Enter patient age: ")
        disease = input("Enter disease: ")

        patient = {
            "Name": name,
            "Age": age,
            "Disease": disease
        }

        patients.append(patient)
        print("Patient added successfully!")

    # View Patients
    elif choice == "2":
        if len(patients) == 0:
            print("No patient records found.")
        else:
            print("\nPatient Records:")
            for i, patient in enumerate(patients, start=1):
                print(f"\nPatient {i}")
                print("Name    :", patient["Name"])
                print("Age     :", patient["Age"])
                print("Disease :", patient["Disease"])
    # Search Patient
    elif choice == "3":
        search_name = input("Enter patient name to search: ")

        found = False

        for patient in patients:
            if patient["Name"].lower() == search_name.lower():
                print("\nPatient Found")
                print("Name    :", patient["Name"])
                print("Age     :", patient["Age"])
                print("Disease :", patient["Disease"])
                found = True
                break

        if not found:
            print("Patient not found.")

    # Remove Patient
    elif choice == "4":
        remove_name = input("Enter patient name to remove: ")

        found = False

        for patient in patients:
            if patient["Name"].lower() == remove_name.lower():
                patients.remove(patient)
                print("Patient removed successfully!")
                found = True
                break

        if not found:
            print("Patient not found.")

    # Exit
    elif choice == "5":
        print("Exiting Hospital Management System...")
        break
    else:
        print("Invalid choice. Please try again.")
```

INPUT & OUTPUT

MAIN MENU

INPUT (User)	OUTPUT (System)
Enter your choice: (User input not required to display menu)	===== HOSPITAL MANAGEMENT SYSTEM ===== 1. Add Patient 2. View Patients 3. Search Patient 4. Remove Patient 5. Exit Enter your choice:

1. ADD PATIENT (First Patient)

INPUT (User)	OUTPUT (System)
Enter your choice: 1 Enter patient name: John Enter patient age: 45 Enter disease: Fever	Patient added successfully!

ADD ANOTHER PATIENT

INPUT (User)	OUTPUT (System)
Enter your choice: 1 Enter patient name: Anita Enter patient age: 30 Enter disease: Diabetes	Patient added successfully!

2. VIEW PATIENTS

INPUT (User)	OUTPUT (System)
Enter your choice: 2	Patient Records: Patient 1 Name : John Age : 45 Disease : Fever Patient 2 Name : Anita Age : 30 Disease : Diabetes

3. SEARCH PATIENT (FOUND)

INPUT (User)	OUTPUT (System)
Enter your choice: 3 Enter patient name to search: John	Patient Found Name : John Age : 45 Disease : Fever

3. SEARCH PATIENT (NOT FOUND)

INPUT (User)	OUTPUT (System)
Enter your choice: 3 Enter patient name to search: Riya	Patient not found.

4. REMOVE PATIENT

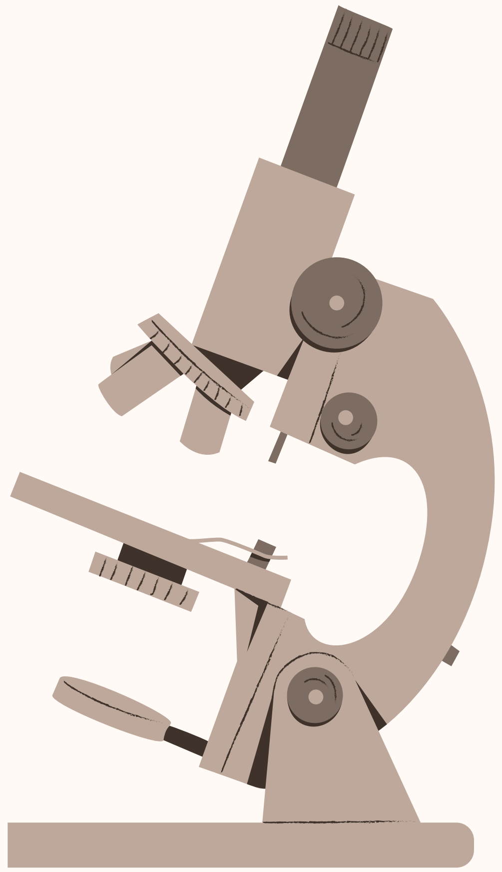
INPUT (User)	OUTPUT (System)
Enter your choice: 4 Enter patient name to remove: Anita	Patient removed successfully!

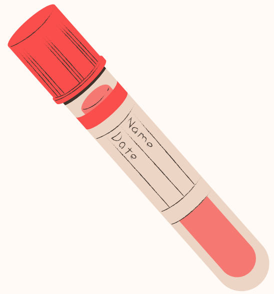
VIEW AFTER REMOVAL

INPUT (User)	OUTPUT (System)
Enter your choice: 2	Patient Records: Patient 1 Name : John Age : 45 Disease : Fever

5. EXIT

INPUT (User)	OUTPUT (System)
Enter your choice: 5	Exiting Hospital Management System...





CONCLUSION

The Hospital Management System (HMS) is an effective software solution that helps hospitals and clinics manage their daily operations efficiently. It simplifies tasks such as patient registration, appointment scheduling, record maintenance, billing, and report generation. By automating these activities, HMS reduces paperwork, saves time, minimizes errors, and improves the quality of healthcare services.

An HMS also enhances communication between different departments like laboratories, pharmacies, and administration, leading to better coordination and faster patient care. Overall, the system improves hospital productivity, ensures secure data management, and provides better satisfaction for both patients and hospital staff.